considering hydrogen – oxygen fuel cell. In this case, hydrogen act as a fuel and oxygen as an oxidant and the electrolyte is aqueous KOH maintained at 200°C and 20 – 40 atm. Porous graphite electrode containing Ni and NiO serves as the inert electrodes.

Hydrogen and oxygen gases are bubbled through the anode and cathode, respectively.

Oxidation occurs at the anode:

$2H_2(g) + 4\,OH^-(aq) \rightarrow 4\,H_2O(l) + 4e^-$

Reduction occurs at the cathode $O_2(g) + 2\,H_2O(l) + 4e^- \rightarrow 4\,OH^-(aq)$

The overall reaction is $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

The above reaction is the same as the hydrogen combustion reaction, however, they do not react directly ie., the oxidation and reduction reactions take place separately at the anode and cathode respectively. Like H_2 - O_2 fuel cell other fuel cells like propane –O_2 and methane O_2 have also been developed.

Corrosion

We are familiar with the rusting of iron. Have you ever noticed a green film formed on copper and brass vessels?. In both, the metal is oxidised by oxygen in presence of moisture. This redox process which causes the deterioration of metal is called corrosion. As the corrosion of iron causes damages to our buildings, bridges etc....it is important to know the chemistry of rusting and how to prevent it. Rusting of iron is an electrochemical process.

Electrochemical mechanism of corrosion

The formation of rust requires both oxygen and water. Since it is an electrochemical redox process, it requires an anode and cathode in different places on the surface of iron. The iron surface and a droplet of water on the surface as shown in figure (9.15) form a tiny galvanic cell. The region enclosed by water is exposed to low amount of oxygen and it acts as the anode. The remaining area has high amount of oxygen and it acts as cathode. So based on the oxygen content, an electro chemical cell is formed. corrosion occurs at the anode i,e,. in the region enclosed by the water as discussed below.

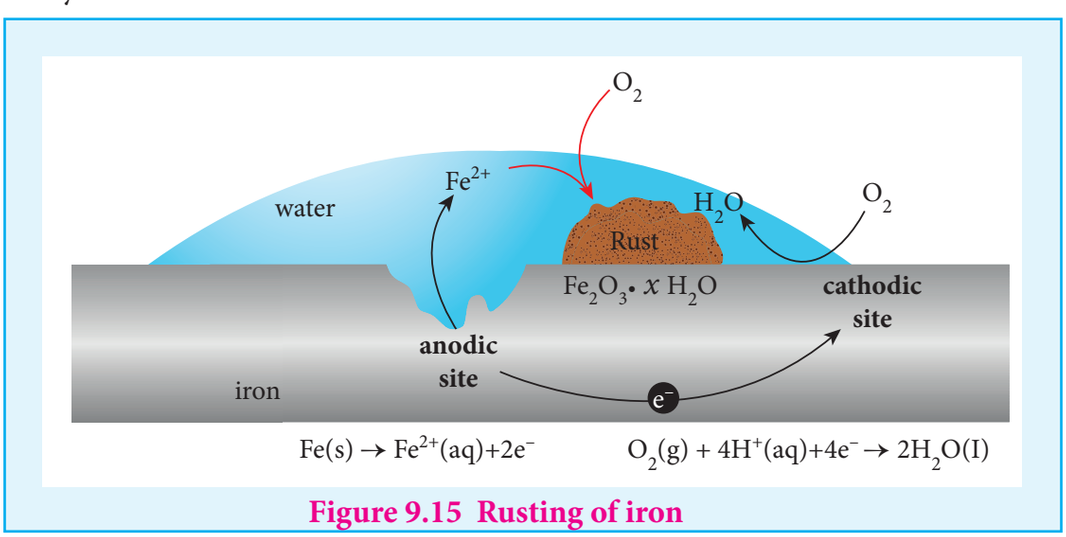


Figure 9.15 Rusting of iron

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